REDUCEDNESS AND SMOOTH MORPHISMS OF ALGEBRAIC SPACES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file spaces-descent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span cites one Morphisms of Spaces lemma, one Descent lemma and one Properties of Spaces section, recorded in assembly.external_references with canonical tags as navigation pointers.

Allowed standard prerequisites (editorial).

- Definition of reduced algebraic spaces. Stacks Project, Properties of Spaces, Section spaces-properties-section-types-properties. Equates reducedness of spaces with reducedness of charts.
- Smoothness is \'etale local on the source. Stacks Project, Morphisms of Spaces, Lemma spaces-morphisms-lemma-smooth-local. Shows $U \to V$ is smooth.
- Reducedness is smooth-local for schemes. Stacks Project, Descent, Lemma descent-lemma-reduced-local-smooth. Gives both directions for $U \to V$.

Source reference. Original reference 03E5: sources/stacks-chow/tags.txt, source lines 4444–4444. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 03ZF: sources/stacks-chow/tags.txt, source lines 5218–5218. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 034E: sources/stacks-chow/tags.txt, source lines 4089–4089. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $f : X \to Y$ be a smooth morphism of algebraic spaces. If Y is reduced, then X is reduced. If f is surjective and X is reduced, then Y is reduced.*

Proof. Choose a commutative diagram

$$\begin{array}{ccc} U & \longrightarrow & V \\ \downarrow & & \downarrow \\ X & \longrightarrow & Y \end{array}$$

where U and V are schemes, the vertical arrows are surjective and étale, and $U \to X \times_Y V$ is surjective étale. Observe that X is a reduced algebraic space if and only if U is a reduced scheme by our definition of reduced algebraic spaces in Properties of Spaces, Section 03E5. Similarly for Y and V. The morphism $U \to V$ is a smooth morphism of schemes, see Morphisms of Spaces, Lemma 03ZF. Since being reduced is local for the smooth topology for schemes (Descent, Lemma 034E) we see that U

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is reduced if V is reduced. On the other hand, if $X \to Y$ is surjective, then $U \to V$ is surjective and in this case if U is reduced, then V is reduced. $\square$